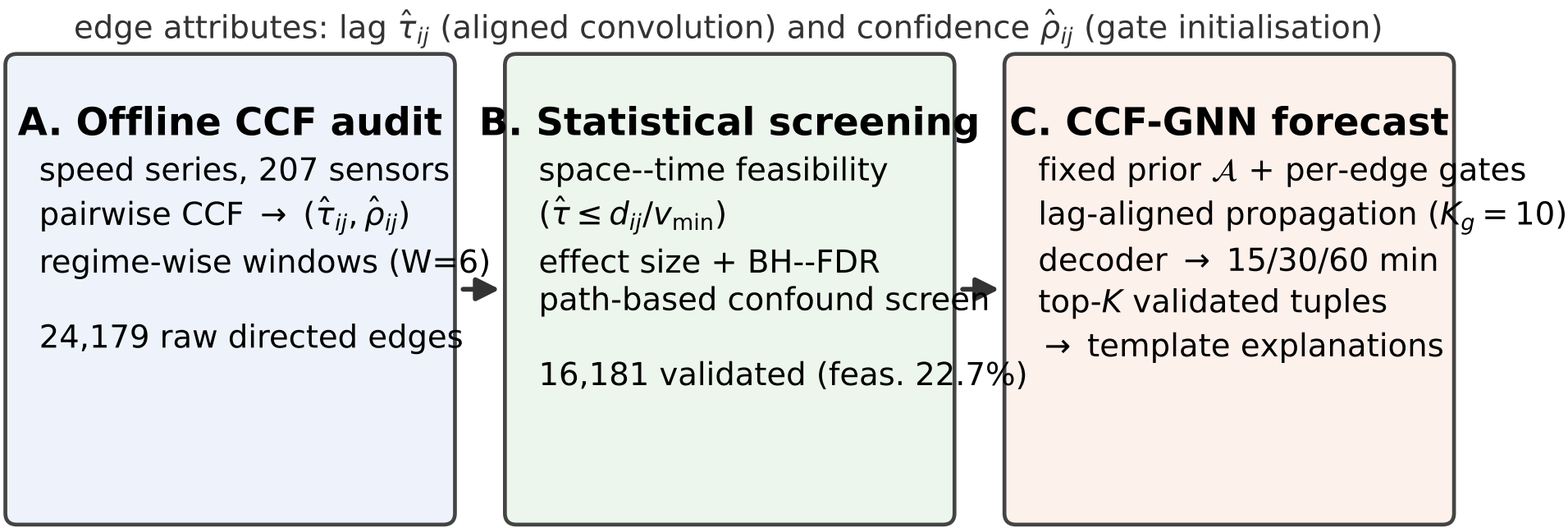
edge attributes: lag $\hat{\tau}_{ij}$ (aligned convolution) and confidence $\hat{\rho}_{ij}$ (gate initialisation)
A. Offline CCF audit
speed series, 207 sensors
pairwise CCF → $(\hat{\tau}_{ij}, \hat{\rho}_{ij})$
regime-wise windows (W=6)
24,179 raw directed edges
B. Statistical screening
space--time feasibility
$(\hat{\tau} \le d_{ij}/v_{\min})$
effect size + BH--FDR
path-based confound screen
16,181 validated (feas. 22.7%)
C. CCF-GNN forecast
fixed prior $\mathcal{A}$ + per-edge gates
lag-aligned propagation ($K_g = 10$)
decoder → 15/30/60 min
top-K validated tuples
→ template explanations
all statistics computed on the training split only; the graph is fixed at test time